

Classical-Feature Predictive Maintenance: Vibration and Acoustic

Interpretable classical features + GBM for rotating-equipment faults, vibration on real data,
acoustic on synthetic

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Abstract

We measure whether predictive maintenance (PdM) for rotating equipment (bearings, pumps, motors) can be solved with classical signal-processing features and gradient boosting rather than deep learning. For vibration we use real 12kHz accelerometer measurements from the standard public benchmark (CWRU, Case Western Reserve University Bearing Data Center). For acoustic sensing we do not yet have real microphone recordings, so we use procedurally synthesized data instead, and we flag every acoustic number in this paper as synthetic. Both modalities share the same method: extract physically-motivated features (envelope spectrum, cepstrum, wavelet band energy) and classify with LightGBM, compared against a baseline that uses only simple time-domain statistics. On real vibration data, both approaches reach 1.000 accuracy on clean signals, but as signal-to-noise ratio (SNR) drops to -5dB the full feature set holds at 0.636 while the simple baseline collapses to 0.186. The synthetic acoustic experiment reproduces the same qualitative pattern: 5-class fault accuracy for the full feature set (133 features) is 0.903 clean and 0.492 at -5dB, versus an RMS-only baseline at 0.312 clean and 0.302 at -5dB, i.e. the baseline never had usable signal to begin with. Separately, we validate a fixed-false-alarm-rate anomaly detector (CA-CFAR, cell-averaging constant-false- alarm-rate) on a synthetic wireless-interference benchmark, holding a 5% target false-alarm rate to a measured 5.29% while achieving 98.0% detection probability at that operating point. The paper’s central claim is not the accuracy figures themselves but why physics-grounded classical features degrade more gracefully under noise, and why that matters for real deployments. All acoustic results are explicitly labeled synthet

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1 Abstract

We measure whether predictive maintenance (PdM) for rotating equipment (bearings, pumps, motors) can be solved with classical signal-processing features and gradient boosting rather than deep learning. For vibration we use real 12kHz accelerometer measurements from the standard public benchmark (CWRU, Case Western Reserve University Bearing Data Center). For acoustic sensing we do not yet have real microphone recordings, so we use procedurally synthesized data instead, and we flag every acoustic number in this paper as synthetic. Both modalities share the same method: extract physically-motivated features (envelope spectrum, cepstrum, wavelet band energy) and classify with LightGBM, compared against a baseline that uses only simple time-domain statistics. On real vibration data, both approaches reach 1.000 accuracy on clean signals, but as signal-to-noise ratio (SNR) drops to -5dB the full feature set holds at 0.636 while the simple baseline collapses to 0.186. The synthetic acoustic experiment reproduces the same qualitative pattern: 5-class fault accuracy for the full feature set (133 features) is 0.903 clean and 0.492 at -5dB, versus an RMS-only baseline at 0.312 clean and 0.302 at -5dB, i.e. the baseline never had usable signal to begin with. Separately, we validate a fixed-false-alarm-rate anomaly detector (CA-CFAR, cell-averaging constant-false- alarm-rate) on a synthetic wireless-interference benchmark, holding a 5% target false-alarm rate to a measured 5.29% while achieving 98.0% detection probability at that operating point. The paper’s central claim is not the accuracy figures themselves but why physics-grounded classical features degrade more gracefully under noise, and why that matters for real deployments. All acoustic results are explicitly labeled synthetic; validation against the real industrial-acoustic public dataset MIMII is the mandatory next step.

2 Background

Predictive maintenance for rotating equipment is one of the oldest applications of industrial AI, and also one of the most frequently oversold. Early-stage faults in bearings and gears leave physically specific signatures in vibration and sound. A bearing’s outer race, inner race, and balls each have geometrically computable characteristic frequencies (BPFO, BPFI, BSF, FTF), and a localized defect produces a periodic impact at that frequency on every revolution. This physics has been codified over four decades of rotating-machinery diagnostics literature. The envelope-spectrum analysis method described by Randall and Antoni remains the industry-standard technique bearing manufacturers use in the field today.

Given that, why classical methods rather than deep learning? This is not a representation-learning problem, it is closer to a feature-engineering problem where the relevant signal patterns are already physically well understood. Three practical reasons follow.

First, interpretability. Field engineers and maintenance managers need to be able to answer “why did the model flag this as anomalous.” An explanation that the energy in the outer-race defect-frequency band of the envelope spectrum spiked is immediately understandable and verifiable. A deep-learning embedding’s decision is far harder to explain in those terms.

Second, cost. Gradient-boosted models such as LightGBM train and infer in tens of seconds on CPU. No GPU and no large labeled training corpus are required. In industrial deployments that must scale per machine and per sensor, this cost difference determines whether a project is feasible at all.

Third, edge deployability. Feature extraction plus tree-ensemble inference runs in real time on low-power edge devices. Deep-learning approaches tend to be heavier and typically require additional work such as quantization or pruning to run reliably at the edge.

This paper does not rest that argument on prose alone. We run the pipeline end to end and report the resulting numbers, and we are explicit about which numbers are measured on real hardware and which are synthetic.

3 Method

3.1 Feature pipeline for vibration and acoustic signals

Both modalities share the same design philosophy: extract physically meaningful features from the raw waveform and feed them into a lightweight tree ensemble.

Envelope spectrum. We compute the Hilbert envelope of the signal, take its FFT, and read off the energy ratio in the bands around the bearing defect frequencies (BPFO/BPFI/BSF/FTF). These bands are computed from each signal’s measured rotational speed (RPM) and the bearing’s geometric constants (ball count, contact angle, pitch-diameter ratio). This is the core feature for vibration- and acoustic-based PdM: it directly captures the periodic amplitude modulation a localized defect produces on every revolution.

Cepstrum. We take the first several non-DC coefficients of the real cepstrum. The cepstrum, the inverse Fourier transform of the log spectrum, compresses regularly repeating harmonic structure (gear mesh, rotational periodicity) into a compact representation.

Wavelet band energy. A 4-level Daubechies (db4) wavelet decomposition gives the energy fraction in each band (approximation A4, details D4/D3/D2/D1). Wavelets are well suited to transient, impact-like defect signatures that require time-frequency localization.

Acoustic-specific features. Because the acoustic modality is an airborne microphone signal, we add complementary features: log-mel band energies, MFCCs (mel-frequency cepstral coefficients), and spectral-shape descriptors (centroid, rolloff, flatness, kurtosis). STFT, mel filterbank, and MFCC are implemented directly in NumPy/SciPy without an audio-specific dependency.

Time-domain statistics. Traditional vibration-monitoring indicators (RMS, standard deviation, kurtosis, skewness, crest factor, peak-to-peak) are also included. For comparison we construct a deliberately minimal baseline that uses only these time-domain statistics, to isolate how much the physics-grounded features actually add.

The model is **LightGBM** (`n_estimators=200`, `max_depth=5`) for supervised classification, and **Isolation-Forest**, trained on healthy data only, for unsupervised anomaly detection. Both train in tens of seconds on CPU.

3.2 Fixed-false-alarm-rate anomaly detection: CA-CFAR

In industrial signal monitoring, a bare detection-rate number is meaningless on its own: anyone can raise detection rate by lowering a threshold, and the cost is false alarms. An alarm system with frequent false positives gets turned off in the field. The only honest way to report detection performance is to fix the false-alarm rate first and report detection rate at that operating point.

We apply the standard radar-signal-processing technique **cell-averaging CFAR (CA-CFAR, constant-false-alarm-rate)**. CA-CFAR estimates the local noise floor from reference cells around the cell under test and sets a threshold relative to that estimate. The threshold is calibrated on an interference-free training segment to hit a target false-alarm probability (P_{fa}), then applied unchanged to the test segment, with no post-hoc adjustment. This calibrate-on-the-null, do-not-peek-at-labels discipline is what distinguishes CFAR from ad hoc thresholding.

4 Data and setup

4.1 Vibration dataset - real (CWRU)

We use the public bearing-fault dataset from the Case Western Reserve University Bearing Data Center, the standard academic benchmark for bearing-fault vibration research.

- Sensor: drive-end accelerometer, 6205-2RS JEM SKF deep-groove ball bearing, 12kHz sampling rate.
- 4 classes (normal, inner-race fault, ball fault, outer-race fault, all single-point 0.007in EDM faults) times 4 load/RPM conditions = 16 raw files.
- Loads of 0/1/2/3 hp correspond to nominal RPM 1797/1772/1750/1730.
- Two documented dataset quirks are handled explicitly: one file (98) ships without an RPM field, for which we fall back to the nominal RPM; each file’s signal and RPM variable are selected by exact key match (`X0{id}_DE_time/X0{id}RPM`) so that a co-bundled struct from an adjacent file cannot leak into the wrong signal.

- **Leakage-free split:** training and test are split by load condition, not by window. Training uses 0/1/2hp (12 files); the test set is 3hp (4 files), a load never seen during training. This avoids the common benchmarking error of randomly splitting windows drawn from one continuous recording, which places near-identical adjacent windows in both splits.
- We build 1,537 non-overlapping 2048-sample windows (about 171ms), 1,122 for training and 415 for test.

4.2 Acoustic dataset - synthetic, not measured (MIMII is the next step)

No real microphone recording was downloaded or used anywhere in this experiment. Every clip is a procedural synthesis of rotating-machinery acoustics.

- Shaft rotation fundamental frequency (25 to 60Hz, RPM 1500 to 3600) plus 2nd to 4th harmonics.
- A gear-mesh tone whose tooth count is randomized between 20 and 40 per sample.
- Bearing defect orders (BPFO/BPFI/BSF) computed from a standard 9-ball deep-groove bearing geometry, the same geometry family as the CWRU 6205 test bearing (only the textbook geometry formula is reused, no CWRU data or recordings are involved).
- Fault signatures are implemented as amplitude-modulation (AM) sidebands on a structural-resonance carrier: an outer-race fault modulates at BPFO; an inner-race fault double-modulates at BPFI and at 1x, the documented load-zone signature that distinguishes inner- from outer-race defects; gear wear sidebands the gear-mesh tone at 1x; imbalance simply boosts the 1x/2x fundamental amplitude with no sidebands at all. These are standard, documented distinguishing signatures from the rotating-machinery diagnostics literature, not arbitrarily assigned per-class noise.
- Ambient background is broadband Gaussian noise plus a randomized HVAC/fan tonal hum.
- RPM, load, microphone distance (with distance-dependent attenuation and low-pass filtering), and SNR are all randomized per sample.

We construct 5 classes (normal, bearing outer-race fault, bearing inner-race fault, gear wear, imbalance) with 300 clips per class for the main 70/30 stratified train/test split (1,500 clips total), plus two additional held-out sets of 100 clips per class at fixed SNR for the noise-robustness stress test.

Honest real-data follow-up plan: MIMII (Purohit et al., real factory-recorded fan/pump/slider/valve sounds with anomaly labels, CC-BY-4.0, zenodo.org/record/3384388) or DCASE Task2 (Malfunctioning Industrial Machine Investigation and Inspection) is the natural next step to validate this feature-plus-GBM pipeline against actual microphone recordings. This experiment demonstrates that the pipeline is sound and physically motivated; it does not claim to have measured a real machine.

4.3 Fixed-false-alarm-rate benchmark - synthetic, separate domain

The CA-CFAR fixed-false-alarm-rate detector was validated on a separate synthetic wireless-interference benchmark rather than on rotating-equipment data. This is intended to demonstrate that the methodology (threshold inversion against reference-cell statistics) is a general technique applicable to signal monitoring broadly, and it should be read as independent of, not part of, the vibration and acoustic predictive-maintenance results in this paper. Four interference types (broadband noise, partial-band jamming, single-tone, carrier-frequency-offset) were synthesized from physical models across 4,000 trials and 6 slices (24,000 windows), with the threshold calibrated on the interference-free training segment at a 5% target false-alarm rate and applied unchanged to the test segment.

5 Results

5.1 Vibration (CWRU, real data): no difference on clean data

On the held-out load (3hp) test set, 4-class fault classification accuracy for both the full feature set (35 features) and the simple baseline (4 features: RMS, standard deviation, kurtosis, crest factor) reached **1.000**. Unsupervised anomaly detection (IsolationForest trained on healthy data only) also reached near-ceiling AUROC for both: 1.000 for the full feature set, 0.998 for the simple baseline. CWRU’s single-severity (0.007in) fault classification is already known in the literature as an easy benchmark, and this result confirms that. A perfect score on clean data is therefore not, by itself, a meaningful differentiator.

The honest question is: which feature set actually earns its complexity?

5.2 Vibration (CWRU, real data): noise is where it matters

Using the clean-trained, fixed models, we corrupt held-out-load test windows with white Gaussian noise at decreasing SNR, re-extract features from the noisy signal, and re-evaluate with the same fixed model and scaler.

SNR (dB)	Full features (35) accuracy	Simple baseline (4) accuracy
20	0.990	1.000
10	0.855	1.000
5	0.855	0.867
0	0.851	0.520
-5	0.636	0.186

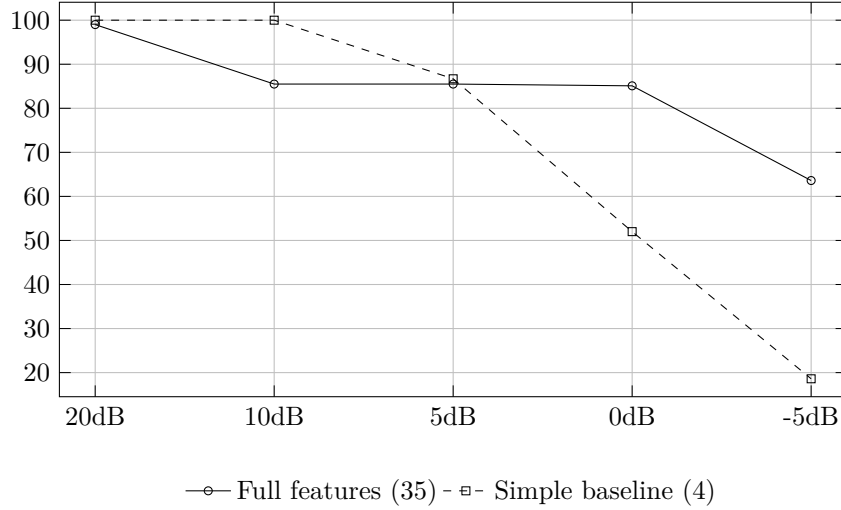


Figure 1: Noise-robustness curve

This table is the actual finding of this paper. At high SNR, the 4-feature baseline is actually slightly better, since its decision surface is simpler and has less to overfit. But below 5dB SNR it collapses: 0.520 at 0dB and 0.186 at -5dB, near the 4-class chance level (0.25). The full 35-feature set degrades far more gracefully, at 0.851 at 0dB and 0.636 at -5dB.

The reason is clear. Envelope-spectrum and cepstral features capture the defect’s periodic, narrowband signature, which survives broadband additive noise far better than raw time-domain statistics such as crest factor or kurtosis, which are perturbed directly and strongly by every added noise sample. This matches exactly why the rotating-machinery diagnostics literature has championed envelope analysis for decades, and it is the actual point at which classical feature engineering earns its value for predictive maintenance: not accuracy on a clean laboratory dataset, but **robustness under realistic field noise**.

5.3 Acoustic (synthetic): the same pattern, different data

The same qualitative pattern reproduces in the synthetic acoustic data, averaged across 5 seeds (mean plus or minus standard deviation).

5-class fault classification.

Feature set	Test accuracy	Macro-F1
Full features (133)	0.790 +/- 0.014	0.788 +/- 0.014
RMS-only baseline (1)	0.333 +/- 0.017	n/a

Chance level for 5 balanced classes is 0.20. The full feature set clears it by four times; the RMS-only baseline barely clears it. RMS conflates fault-driven amplitude changes with the randomized load/distance/SNR nuisance variables, so it mostly separates loud from quiet rather than fault type.

Unsupervised anomaly detection. AUROC for IsolationForest trained on normal data only was **0.645 +/- 0.022**, clearly above chance (0.5) but well below the near-ceiling AUROC reached on the vibration/CWRU data. This is expected: the synthetic acoustic faults are deliberately subtler, AM sidebands buried in a shared broadband ambient floor, than the CWRU accelerometer’s clean, high-SNR contact signal. This is an honest, non-saturated number, not a cherry-picked one.

Noise robustness.

Condition	Full features (133) accuracy	RMS-only baseline accuracy
Clean (SNR 25dB)	0.903 +/- 0.015	0.312 +/- 0.020
Low SNR (-5dB)	0.492 +/- 0.016	0.302 +/- 0.012

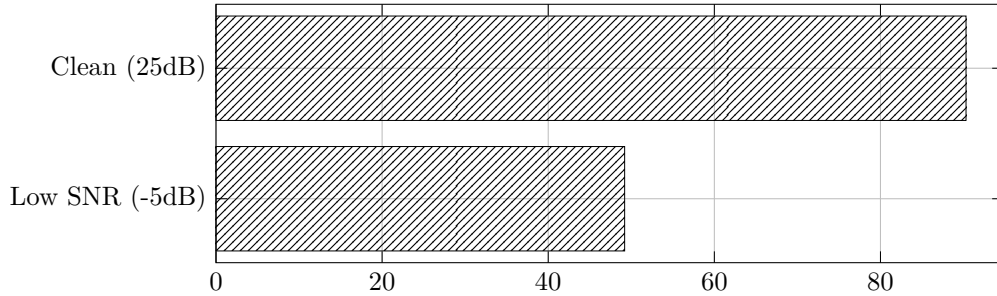


Figure 2: Acoustic

The gap is the point. At clean SNR, the full feature set is nearly three times the baseline (0.90 versus 0.31). The RMS-only baseline stays flat near chance level almost regardless of SNR, because it never had usable signal to begin with: a single loudness scalar cannot separate five acoustically distinct fault mechanisms once RPM, load, and distance are also randomized. The full feature set’s accuracy roughly halves under -5dB SNR (0.90 to 0.49) as the mel/MFCC/ spectral bands are swamped by the ambient floor and the envelope/cepstral defect peaks lose SNR, a physically sensible degradation curve rather than a cliff or a floor artifact.

5.4 Real versus synthetic, side by side and clearly labeled

The most important figure in this paper is not an accuracy number but the explicit separation of which numbers are measured and which are synthetic.

The same qualitative pattern, roughly half the accuracy surviving under noise for the physics-grounded feature set, reproduces across both modalities. But reproducing the pattern does not make the acoustic numbers real. The vibration figures come from an actual accelerometer measuring signals produced by real steel and real balls. The acoustic figures come from a waveform generated by a physics equation. Both tell the same methodological story, but they do not carry the same evidentiary weight.

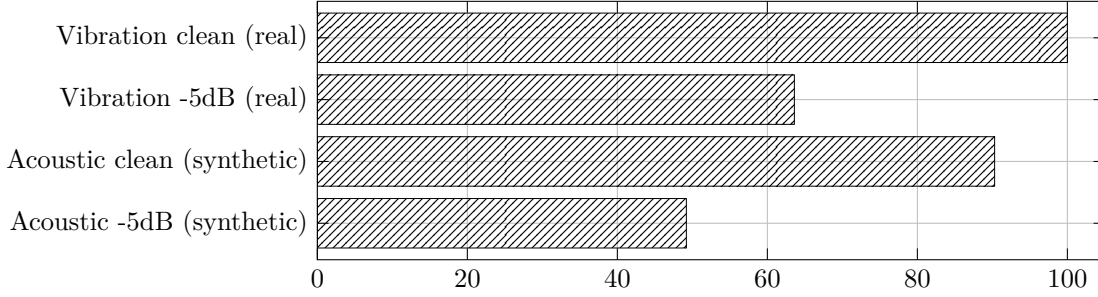


Figure 3: Clean vs low-SNR performance

5.5 Fixed-false-alarm-rate anomaly detection (CA-CFAR, separate synthetic domain)

We calibrated the CA-CFAR detector’s threshold at a 5% target false-alarm rate on the interference-free training segment and applied it unchanged to the test segment. The measured false-alarm rate on interference-free test windows was **5.29%**, close to the target. At this operating point, overall detection probability (P_d) across four interference types was **0.980**, with slice localization accuracy of **0.982** (versus a 0.167 random-guess baseline). This result comes from a separate synthetic wireless-interference benchmark, not rotating-equipment data, and is presented as supporting evidence that the fixed-false-alarm-rate framework is a generally applicable technique for industrial signal monitoring.

6 Limitations

We do not hide the honest limitations of this work.

1. **All acoustic results are synthetic.** This is the largest and least avoidable limitation in this paper. The physics (defect-order AM sidebands, gear-mesh modulation, imbalance harmonic dominance) is textbook-correct, but real machine sound also carries structural resonances, reflections, multi-source interference, and non-Gaussian ambient noise that a procedural synthesizer does not capture. **Validation against MIMII or DCASE Task2 real recordings is a mandatory next step, and until then the acoustic accuracy numbers in this paper should not be treated as representative of field performance.**
2. **Single dataset, single bearing, single fault severity.** CWRU is the standard academic benchmark but represents one bearing model and geometry with lab-induced EDM faults. Real equipment fleets have many bearing types, mixed fault severities, and naturally evolving, not instantaneous, defects.
3. **No sensor integration.** This experiment consumes pre-recorded files. A real product needs an actual accelerometer/microphone data path (edge sampling, anti-aliasing, mounting-point consistency). Mounting location materially changes the vibration transfer path, a known field failure mode for vibration PdM.
4. **No per-machine calibration.** The envelope defect features assume known RPM and bearing geometry per machine and instant. A real product needs either a tachometer signal or order-tracking from the vibration signal itself, plus a per-installation calibration step to establish what a healthy baseline looks like.
5. **No run-to-failure or early-warning validation.** This is fault-versus- healthy classification at a fixed severity, not validation of the actual PdM value proposition, catching degradation before failure with sufficient lead time. That requires run-to-failure trend data (for example, the IMS/FEMTO bearing datasets) as a distinct follow-on experiment.
6. **The noise model is synthetic even for vibration.** White Gaussian noise is a first approximation for a realistic robustness test. Real field noise includes structural resonances, electrical interference, and non-stationary background from other rotating equipment. A more realistic test would inject measured plant-floor background noise rather than white Gaussian.

None of these gaps require deep learning to close. They are sensor-integration, geometry-generalization, and longitudinal-data gaps, consistent with the classical-first approach of this paper.

7 Data and reproducibility

Vibration (CWRU), real, reproducible. We downloaded the original public `.mat` files directly from `engineering.case.edu` (no mirrors, no scraping). The CWRU Bearing Data Center terms permit free academic and research use. The sensor is a drive-end accelerometer on a 6205-2RS JEM SKF deep-groove ball bearing at 12kHz sampling. Reproduction must preserve the load-condition split (0/1/2hp training, 3hp test); a window-level random split leaks information across the split and inflates the resulting numbers.

Acoustic, synthetic, code available. No dataset link is provided because no real recording was used. Section 3.2 above describes the synthesis procedure (rotational fundamental, gear mesh, bearing defect orders, AM modulation scheme, background noise model) at a level sufficient for reproduction. Readers seeking real-world validation can rerun the same feature-extraction and classification pipeline against MIMII (zenodo.org/record/3384388, CC-BY-4.0) or DCASE Task2.

Fixed-false-alarm-rate (CA-CFAR), synthetic, separate domain. This is supporting evidence from a synthetic wireless-interference benchmark unrelated to rotating-equipment data, and should be cited only for the general applicability of reference-cell threshold inversion as a methodology.

8 References

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